

human

Where are the car parks in Exeter?

host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': 'User wants to find the locations of car parks in Exeter. Provide the general steps to solve this geospatial query.'}

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entity to search for is car parks, and we need to find as many entries as possible for car parks in Exeter.

output steps: ["Find Exeter 1 area", "Search for car parks in Exeter as many search results"]

host_agent

Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}

Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..,\n count of records fetched : 98.\n ',]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific area polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns for high quality
    filtered = gdf[
        (
            (gdf['name1_text'].str.lower() == 'exeter') |
            (gdf['name1_text'].str.lower().str.contains('exeter')) |
            (gdf['sameasdbpedia'].str.lower().str.contains('exeter', na=False)) |
            (gdf['sameasgeonames'].str.lower().str.contains('2649808', na=False)) # 2649808 is Exeter's GeoNames ID
        ) & (
            (gdf['descriptiongroup'].str.lower().isin(['settlement', 'part of settlement'])) |
            (gdf['description'].str.lower().isin(['city', 'town', 'suburban area']))
        )
    ]
    return [
        f"Filtered to {len(filtered)} polygon(s) for Exeter area.",
        'filtered_exeter_area_polygon',
        'GeoDataFrame containing only the polygon(s) representing the Exeter area, filtered using name, dbpedia, geonames, and description columns for high accuracy.',
        filtered
    ]
```

Filtered to 2 polygon(s) for Exeter area. - Artifact name: filtered_exeter_area_polygon - Description: GeoDataFrame containing only the polygon(s) representing the Exeter area, filtered using name, dbpedia, geonames, and description columns for high accuracy.

named_area_agent

The specific area polygon for Exeter has been filtered and saved. Here are the details:

- **Artifact Name:** filtered_exeter_area_polygon
- **Description:** GeoDataFrame containing only the polygon(s) representing the Exeter area, filtered using name, dbpedia, geonames, and description columns for high accuracy.

host_agent

Calling tool send_message with args : {'target': 'land_use_features', 'task_description': "Search for car parks within the area defined by the artifact 'filtered_exeter_area_polygon'."}

land_use_features_agent

Calling tool call_os_ngd with args : {'filters': ['Public Car And Coach Park', 'Public Car And Commercial Vehicle Park', 'Public Car Park'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'car_parks_in_exeter'}

Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : ['landuse_car_parks_in_exeter'],\n Descriptions are : ["A geopandas dataframe containing land use data with filters and bbox applied for the query Search for car parks within the area defined by the artifact 'filtered_exeter_area_polygon'. using filters ['Public Car And Coach Park', 'Public Car And Commercial Vehicle Park', 'Public Car Park']. Further Name filtering is available for this"],\n counts of records fetched are : [45].\n ', []]

I found 45 records of car parks within the area defined by 'filtered_exeter_area_polygon'. The data is saved in the artifact named 'landuse_car_parks_in_exeter'.

If you need further analysis or filtering, please let me know!

host_agent

Calling tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot the locations of all car parks from the artifact 'landuse_car_parks_in_exeter' within the area of 'filtered_exeter_area_polygon'."}

plotting_agent

Tool Code Executor

```
def plot_car_parks_within_exeter_area(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import Polygon

    # Unpack data
    car_parks = data[0]
    exeter_area = data[1]

    # Ensure both are in the same CRS (EPSG:4326 for folium)
    car_parks = car_parks.to_crs(epsg=4326)
    exeter_area = exeter_area.to_crs(epsg=4326)

    # Select the Exeter city polygon (description == 'City' and name1_text == 'Exeter')
    exeter_city_poly = exeter_area[(exeter_area['description'] == 'City') & (exeter_area['name1_text'] == 'Exeter')]
    if exeter_city_poly.empty:
        return ["No Exeter city polygon found in the area data.", None, None, None]
```

```

# Spatial join: car parks within Exeter city polygon
car_parks_in_exeter = gpd.sjoin(car_parks, exeter_city_poly, predicate='within')

# Create folium map centered on Exeter
centroid = exeter_city_poly.geometry.iloc[0].centroid
fmap = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

# Plot Exeter city boundary
folium.GeoJson(exeter_city_poly.geometry.iloc[0], name='Exeter City Boundary', style_function=lambda x: {'color': 'blue', 'fill': False, 'weight': 2}).add_to(fmap)

# Plot car parks
for _, row in car_parks_in_exeter.iterrows():
    folium.GeoJson(row['geometry'], name='Car Park', style_function=lambda x: {'color': 'red', 'fill': True, 'fillOpacity': 0.5, 'weight': 1}).add_to(fmap)
    # Add popup with name if available
    popup_text = row['name1_text'] if row['name1_text'] else 'Car Park'
    folium.Popup(popup_text).add_to(folium.GeoJson(row['geometry']))

# Save map
map_filename = 'car_parks_within_exeter_area.html'
fmap.save(map_filename)

# Prepare summary
summary = f"Plotted {len(car_parks_in_exeter)} car parks within the Exeter city area. First 5 car parks: "
if len(car_parks_in_exeter) > 0:
    summary += str(car_parks_in_exeter[['osid', 'name1_text', 'geometry_area_m2']].head(5).to_dict(orient='records'))
else:
    summary += "No car parks found within the Exeter city area."

artifact_description = "Folium map showing all car parks (in red) within the Exeter city area boundary (in blue). Includes popups for car park names if available."
return [summary, 'car_parks_within_exeter_area_map', artifact_description, map_filename]

plot_car_parks_within_exeter_area

```

Tool Code Executor

```

def plot_car_parks_within_exeter_area(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import Polygon

    # Unpack data
    car_parks = data[0]
    exeter_area = data[1]

    # Ensure both are in the same CRS (EPSG:4326 for folium)
    car_parks = car_parks.to_crs(epsg=4326)

```

```

exeter_area = exeter_area.to_crs(epsg=4326)

# Check available columns for exeter_area
exeter_area_cols = exeter_area.columns
# Use correct column for name (from metadata: 'name1_text')
# Select the Exeter city polygon (description == 'City')
exeter_city_poly = exeter_area[(exeter_area['description'] == 'City')]
if exeter_city_poly.empty:
    return ["No Exeter city polygon found in the area data.", None, None, None]

# Spatial join: car parks within Exeter city polygon
car_parks_in_exeter = gpd.sjoin(car_parks, exeter_city_poly, predicate='within')

# Create folium map centered on Exeter
centroid = exeter_city_poly.geometry.iloc[0].centroid
fmap = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

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# Save map
map_filename = 'car_parks_within_exeter_area.html'
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    summary += str(car_parks_in_exeter[['osid', 'geometry_area_m2']].head(5).to_dict(orient='records'))
else:
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artifact_description = "Folium map showing all car parks (in red) within the Exeter city area boundary (in blue)."
```

return [summary, 'car_parks_within_exeter_area_map', artifact_description, map_filename]

plot_car_parks_within_exeter_area

Tool Code Executor

```

def plot_car_parks_within_exeter_area(data: list):
    import geopandas as gpd
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```

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car_parks = car_parks.to_crs(epsg=4326)
exeter_area = exeter_area.to_crs(epsg=4326)

# Select the Exeter city polygon (description == 'City')
exeter_city_poly = exeter_area[exeter_area['description'] == 'City']
if exeter_city_poly.empty:
    return ["No Exeter city polygon found in the area data.", None, None, None]

# Spatial join: car parks within Exeter city polygon
car_parks_in_exeter = gpd.sjoin(car_parks, exeter_city_poly, predicate='within')

# Create folium map centered on Exeter
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# Plot car parks
for _, row in car_parks_in_exeter.iterrows():
    folium.GeoJson(row['geometry'], name='Car Park', style_function=lambda x: {'color': 'red', 'fill': True, 'fillOpacity': 0.5, 'weight': 1}).add_to(fmap)

# Save map
map_filename = 'car_parks_within_exeter_area.html'
fmap.save(map_filename)

# Prepare summary using only columns that exist
available_cols = car_parks_in_exeter.columns
summary_cols = [col for col in ['geometry_area_m2', 'description'] if col in available_cols]
summary = f"Plotted {len(car_parks_in_exeter)} car parks within the Exeter city area. First 5 car parks: "
if len(car_parks_in_exeter) > 0:
    summary += str(car_parks_in_exeter[summary_cols].head(5).to_dict(orient='records'))
else:
    summary += "No car parks found within the Exeter city area."

artifact_description = "Folium map showing all car parks (in red) within the Exeter city area boundary (in blue)."
```

`return [summary, 'car_parks_within_exeter_area_map', artifact_description, map_filename]`

`plot_car_parks_within_exeter_area`

Plotted 44 car parks within the Exeter city area. First 5 car parks: [{'geometry_area_m2': 13954.243}, {'geometry_area_m2': 2409.06}, {'geometry_area_m2': 2346.912}, {'geometry_area_m2': 1153.566}, {'geometry_area_m2': 3032.792}]

Artifact name: car_parks_within_exeter_area_map

Artifact description: Folium map showing all car parks (in red) within the Exeter city area boundary (in blue).

Artifact data: car_parks_within_exeter_area.html

host_agent

The car parks in Exeter have been located and mapped. There are 44 car parks identified within the Exeter city area.

- The locations were found using the official land use database, filtered specifically for car parks within the accurate boundary of Exeter.
- A map has been generated showing all car parks (in red) within the Exeter city area boundary (in blue).

If you would like to view the map or need further details about specific car parks, please let me know!